

Warm Worked Low Carbon Co-28Cr-6Mo Alloy

Warm worked R31537 cobalt is R31537 cobalt in the warm worked condition. It has the highest strength compared to the other variants of R31537 cobalt. The graph bars on the material properties cards below compare warm worked R31537 cobalt to other cobalt alloys (top) and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

Mechanical Properties

Elastic (Young's, Tensile) Modulus

220 GPa 32×10^6 psi

Elongation at Break

14 %

Fatigue Strength

480 MPa 70×10^3 psi

Poisson's Ratio

0.29

Reduction in Area

14 %

Rockwell C Hardness

35

Shear Modulus

87 GPa 13×10^6 psi

Tensile Strength: Ultimate (UTS)

1340 MPa 190×10^3 psi

Tensile Strength: Yield (Proof)

940 MPa 140×10^3 psi

Thermal Properties

Latent Heat of Fusion

320 J/g

Melting Completion (Liquidus)

1360 °C2480 °F

Melting Onset (Solidus)

1290 °C2360 °F

Specific Heat Capacity

450 J/kg-K0.11 BTU/lb-°F

Thermal Conductivity

13 W/m-K7.3 BTU/h-ft-°F

Thermal Expansion

13 µm/m-K

Electrical Properties

Electrical Conductivity: Equal Volume

2.0 % IACS

Electrical Conductivity: Equal Weight (Specific)

2.1 % IACS

Otherwise Unclassified Properties

Density

8.4 g/cm³530 lb/ft³

Embodied Carbon

8.1 kg CO₂/kg material

Embodied Energy

110 MJ/kg46 x 10³ BTU/lb

Embodied Water

530 L/kg63 gal/lb

Common Calculations

Resilience: Ultimate (Unit Rupture Work)

160 MJ/m³

Resilience: Unit (Modulus of Resilience)

1990 kJ/m³

Stiffness to Weight: Axial

15 points

Stiffness to Weight: Bending

24 points

Strength to Weight: Axial

44 points

Strength to Weight: Bending

32 points

Thermal Diffusivity

3.4 mm²/s

Thermal Shock Resistance

32 points